



National University of Computer and Emerging Sciences

Department		Computer Science		Dept. Code		CS	
Course Title		Technical and Business Writing		Course Code		SS2012	
Pre-requisite (s)		SS1014		Credit Hrs.		3	
Course Objective:		The purpose of this course is to enable students to understand the definition and the style of technical communication. The students will learn how to produce effective technical documents, like, reports, user manuals, specification, etc. in business and industry. They will learn the universally accepted and international standards of technical communication. Using principles of analyzing and planning to meet the reader’s informational needs, students produce proposals, instructions and the various types of informative and persuasive reports used in organizations. In this way, they will develop skills necessary for effective performance in professional life.					
PLO		Program Learning Outcome (PLO) Statement					
10		Communication		Communicate effectively on complex computing activities with the computing community and with society at large.			
12		Life-long Learning		Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.			
CLO	Course Learning Outcome (CLO)		Domain	Taxonomy Level	PLO	Tools	
01	Utilize efficient writing style for producing an effective technical document.		Cognitive	3	10	A, M, Q	
02	Compose reports for effective performance in professional life.		Cognitive	6	10	A, F, CA, Q	
03	Design, document, and develop a research project.		Cognitive	6	12	RP, M, F, Q	
		<i>Tool: A = Assignment, M = Midterm, F=Final, CA =Class Activity ,RP = Research Project</i>					
Text Book(s)		Title		Technical Communication and its applications			
		Author		Jerome N. Borowick			
		Publisher		Prentice Hall			
Ref. Book(s)		Title		Technical Writing			
		Author		John M. Lannon			
		Publisher		Scott Foresman & Co.			
		Title		Writing for Computer Science			
		Author		Justin Zobel.			
		Publisher		Springer.			

Assessments	% Marks		
1. Assignments	2+2+3+2		
2. Quizzes	2+2	5. Mid-Terms	30 %
3. Final Project	5+2	6. Final Exam	50 %
Total:-			100

Weeks	Contents/Topics	CLOs	AI Tools for Students (Guided & Ethical Use)	Assessment Tools
01	- Orientation class - Technical Writing: Definitions, History, Purposes, Functions, Defining Characteristics	1	ChatGPT (definitions & examples), Gemini (comparison of definitions)	A1, M1, Q1
02	The Technical Style: Clarity, Precision, Objectivity, Simplicity, & Economy	1	Hemingway Editor, Grammarly Free (clarity & conciseness checks)	A1, M1, Q1
03	- The Technical Writing Process- Purpose analysis & Audience Analysis - Data Collection & Analysis- Primary & Sources, Qualitative & Quantitative Data Quiz 1 (Topics of week 1 & 2)	1	ChatGPT (audience analysis prompts), Google Forms (survey drafts)	A2, M1
04	- CV/Resume Writing - Cover Letters Assignment 1: Resume writing	2	Zotero, Scite.ai, ChatGPT (idea integration, not writing)	A1, M1
05	How to write instructions in user guides Assignment 2: Making a User guide	2	Notion AI, Canva Docs, ChatGPT (step ordering & clarity)	A2, F
06	MID 1			
07	- How to read a Research Paper - Introduction to Scientific Research - Research Proposal Project: Writing Research Proposal	3	Resume Worded, Grammarly, ChatGPT (bullet improvement)	A3, M2
08	- Constructing Effective Paragraphs for the technical prose - Writing Synthesis Essay - Introduction to Citation and Referencing Class Assignment 3: Synthesis	1	Elicit.org, Research Rabbit, ChatGPT (proposal outlines)	FP, M2, Q2
09	The Technical Report: Writing the Introduction and Literature Review Sections	3	Connected Papers, Scite.ai, ChatGPT (gap identification)	FP, M2, F, Q2
10	Technical Reports: Method, Results, Conclusion and Recommendation Sections	3	Grammarly, ChatGPT (section differentiation & tone)	FP, M2, Q2
11	Quiz 2 (Topics of week 9 & 10) - Preparing Prefatory Parts for Technical Reports: Title Page, Table of Contents, Letter of Transmittal, Abstract, & Executive Summary	3	Canva Docs, ChatGPT (abstract vs executive summary)	FP, M2
12	MID-II			
13	- Short Survey Report - Progress Reports	2	Google Forms, Excel AI, ChatGPT (data interpretation)	F
14	- Feasibility Studies - Technical Proposal Quiz 3 (Topics of week 13)	2	Notion AI, ChatGPT (planning & structuring reports)	F, CP
15	Professional Emails Class Assignment 4 (Feasibility Studies)	2	Grammarly, Outlook Copilot, ChatGPT (tone & clarity)	F, CP
16	Project Presentation	3	Canva, Gamma AI, ChatGPT (presentation rehearsal)	F, CP

Responsible Use of AI Tools in Technical & Business Writing (SS2007)

This course allows guided and ethical use of Artificial Intelligence (AI) tools to support learning, clarity, and revision. AI is a learning assistant, not a replacement for student thinking or writing.

Permitted Uses

Students may use AI tools for:

- Brainstorming ideas and outlines
- Understanding writing formats (reports, proposals, emails, resumes)
- Improving clarity, grammar, coherence, and tone
- Summarizing sources after reading them independently
- Checking structure and logical flow
- Preparing presentations and visual aids

Prohibited Uses

Students may NOT:

- Submit AI-generated text as their own work
- Use AI to write full assignments, reports, or essays
- Use AI during quizzes, midterms, or exams
- Bypass reading, research, or drafting processes

Academic Integrity

- All submitted work must reflect the student's own ideas, organization, and voice
- Excessive or unethical AI use may result in penalties under university plagiarism policies
- Students may be asked to explain or defend their written work orally

Rule of Thumb:

If AI helps you think → allowed

If AI thinks for you → not allowed